

Yash Kherwal

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EDUCATION

Indian Institute of Technology Roorkee

2022 – 2026

B.Tech in Chemical Engineering

EXPERIENCE

AI Product Manager Intern

Feb 2026 – Apr 2026

Orange Labs — EdTech ERP platform for schools and colleges

- Built **CCTV-based automated attendance pipeline** — computer vision model detects and identifies students from live camera feeds, auto-syncing presence records into the school ERP; eliminated manual marking and enabled real-time classroom tracking.
- Engineered **AI-generated personalised lecture pipeline**: ingested real teacher audio/video as training data to synthesise **teacher-cloned video lectures** — students receive curriculum content delivered in their own school teacher's voice and likeness, improving engagement over generic video content.

Software Development Intern

May 2025 – June 2025

ScoreMe Solutions

- Designed a **two-stage ML classification pipeline** for fintech PDF processing — rule-based heuristics handle deterministic inputs first; **Random Forest classifier** resolves ambiguous cases, avoiding unnecessary model inference on clear-cut documents and reducing classification latency.
- Implemented **confidence calibration layer**: predictions below threshold withheld from downstream **credit scoring pipelines** and flagged for analyst review — prevents silent misclassification propagation; integrated **Tesseract OCR** via **PDFBox** as fallback for scanned inputs, extending pipeline coverage.

Engineering Intern

June 2025 – July 2025

Bharat Electronics Limited

- Implemented a **binary stream parser** for **ASTERIX CAT048** radar protocol per EUROCONTROL spec — decoded variable-length optional-field packet structures into normalised aviation records; designed **concurrent ingestion** with **backpressure handling** to maintain throughput under sustained high-frequency surveillance loads.

PROJECTS

CareerAutomated | Python, LangGraph, Groq (LLaMA), SQLite, Pandas, Playwright, IMAP/SMTP, FastAPI 2025

- Built **autonomous AI recruiting platform** orchestrating job discovery, candidate-fit scoring, resume tailoring, recruiter outreach, **ATS browser automation**, CRM tracking, and inbox intelligence through a **unified multi-agent workflow**.
- Designed **Generator-Critic agent architecture**: specialised agents perform company intelligence, project selection, and personalised email generation; **Critic agent** validates quality, contextual relevance, formatting, and placeholder safety before any outreach is dispatched.
- Engineered **Contact State Engine** via **IMAP Gmail sync + SQLite CRM** — deduplication pipeline removes malformed records and previously-contacted recruiters across **10,000+ contacts**; **Playwright** automates ATS form execution with per-platform handlers and checkpoint recovery.

Hybrid RAG Retrieval System | Python, LangChain, BGE-M3, AstraDB, FastAPI, AWS EC2 2024

- Diagnosed keyword retrieval failure on 500+ document corpus — BM25 missed semantically relevant results due to vocabulary mismatch; introduced **hybrid retrieval** combining **BGE-M3** dense embeddings with sparse BM25 scores, improving recall on paraphrased and domain-specific queries.
- Designed **decoupled ingestion and serving**: offline embedding pipeline feeds **AstraDB** independently of **FastAPI** serving layer on **AWS EC2** — corpus updates without downtime; **response groundedness** enforced via prompt constraints to prevent hallucination on out-of-context queries.

YAAR — Personality-Adaptive LLM System | React Native, LLM APIs, FastAPI, Persistent Memory 2025

- Designed a **persistent personality memory system**: each interaction updates a **personality metric vector** (humour tolerance, self-awareness index, reaction intensity) injected into the **LLM context window** at inference time to steer tone without fine-tuning; shipped **Android APK** via React Native + EAS Build.

TECHNICAL SKILLS

LLM & Agentic AI: LangGraph, LangChain, Multi-Agent Orchestration, Generator-Critic Patterns, MCP, Structured Outputs, Prompt Engineering

Computer Vision: Object Detection, Face Recognition, Real-Time Video Processing, OpenCV, CCTV Analytics

Retrieval: Hybrid RAG, BGE-M3, Dense + Sparse Retrieval, AstraDB, Vector Databases, ANN Indexing

ML / AI: TensorFlow, Scikit-learn, Random Forest, Confidence Calibration, Inference Optimization

Backend & Infra: FastAPI, Docker, AWS EC2, SQLite, Playwright, IMAP/SMTP, REST APIs, Stream Processing

Languages & Data: Python, SQL, Java, C++, DSA, Pandas, pdfplumber